Anisotropic Azimuthal Skin Funneling, HONHSE, & Non-Bloch Band Topology in Chiral Icosahedral Quasicrystals

From 6D Cut-and-Project Lattices to Multi-Messenger Cosmology

Mathematical Physics & Quasicrystal Research Program

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Slide 1: 6D Hypercubic Cut-and-Project & Theorem 3.1

- **Superspace Decomposition:** $\mathbb{R}^6 = E_\parallel \oplus E_\perp$ via golden ratio $\tau = \frac{1+\sqrt{5}}{2}$ projectors $\mathcal{P}_\parallel, \mathcal{P}_\perp$.
- **Theorem 3.1 (Verified):** Exact completeness $\sum_{\alpha=1}^{6} v_{\alpha,a} v_{\alpha,b} = \delta_{ab}$, $\|\boldsymbol{v}_\alpha\|^2 = \|\boldsymbol{w}_\alpha\|^2 = 1/2$, and Galois conjugation ($\tau \mapsto -1/\tau$) between physical star vectors $\boldsymbol{v}_\alpha \in E_\parallel$ and phason star vectors $\boldsymbol{w}_\alpha \in E_\perp$.
- **Triacontahedral Window $\mathcal{W}$:** 30 facets uniquely oriented by $\binom{6}{2} = 15$ normal pairs $\hat{\boldsymbol{n}}^{\pm}_{(\alpha\beta)} = \pm \frac{\boldsymbol{w}_\alpha \times \boldsymbol{w}_\beta}{\|\boldsymbol{w}_\alpha \times \boldsymbol{w}_\beta\|}$.
- **Thermodynamic Scaling:** Cluster size $N_{\text{sites}} = \mathcal{O}(L^3)$ preserves exact I_h point-group symmetry.

Slide 2: Non-Unitary Q_8 Transport & Azimuthal Skin Funneling

- **Quaternion Edge Operators:** Directed tight-binding transport $\mathcal{T}_{ij} = t_0 \exp(\gamma \hat{\boldsymbol{d}}_{ij} \cdot \boldsymbol{\Omega}_{\text{chiral}} + i\alpha \hat{\boldsymbol{d}}_{ij} \cdot \boldsymbol{\sigma})$ maps $Q_8 \to \mathrm{SU}(2)$ spinor hops.
- **Non-Conservative Curl:** $\nabla \times \boldsymbol{\Omega}_{\text{chiral}} \neq \mathbf{0}$ around the 12 five-fold disclination axes prevents imaginary gauge elimination under OBC.
- **Azimuthal Funneling:** Bulk wavepackets spiral transversely across rhombic faces and funnel into the 12 five-fold icosahedral vertices (6.74× mode gain).

Slide 3: Codimension-2/3 HONHSE & Corner Localization

- **Higher-Order Skin Effect (HONHSE):** Macroscopic $\mathcal{O}(L^3)$ bulk states collapse onto $\mathcal{O}(1)$ icosahedral corners rather than $\mathcal{O}(L^2)$ surfaces.
- **Inverse Participation Ratio (IPR):**

$$\mathrm{IPR}_n = \sum_i |\psi_{n,R}(i)|^4 \sim \mathcal{O}(L^0) \quad \text{(OBC)} \quad \text{vs.} \quad \mathcal{O}(L^{-3}) \quad \text{(PBC Approximant)}.$$

- **Biorthogonal Delocalization:** While right eigenstates ψ_R pile up at the 12 vertices, $\rho_{\mathrm{bio}}(i) = |\psi_L^*(i)\psi_R(i)|$ remains bulk-distributed.

Slide 4: Non-Bloch Band Topology, GBZ Scaling, & Phason Loops

- **Surrogate 6D GBZ:** Complex momentum lift $\boldsymbol{K} = \boldsymbol{k} + i\boldsymbol{\kappa} \in \mathbb{C}^6$ defines the Generalized Brillouin Zone $\mathcal{M}_{\mathrm{GBZ}}$ via $|\beta_M(\boldsymbol{\Phi})| = |\beta_{M+1}(\boldsymbol{\Phi})|$ ($\Delta\kappa(W_b) = +1.51898 \times 10^{-2}$).
- **Phason Point-Gap Winding:** Adiabatic phason flips $\boldsymbol{\Phi}(t) \in \mathcal{C}_\perp \subset E_\perp$ wind the complex energy spectrum around E_0:

$$W(E_0) = \oint_{\mathcal{C}_\perp} \frac{d\Phi_\mu}{2\pi i} \partial_{\Phi_\mu} \ln \det[H(\boldsymbol{\Phi}) - E_0] \in \mathbb{Z}.$$

Slide 5: Quantum Geometric Tensor (QGT) & Floquet Phason Drive

- **Biorthogonal QGT:** $\mathcal{Q}_{\mu\nu}^{(n)} = \langle \partial_\mu \psi_n^L | \partial_\nu \psi_n^R \rangle - \langle \partial_\mu \psi_n^L | \psi_n^R \rangle \langle \psi_n^L | \partial_\nu \psi_n^R \rangle = g_{\mu\nu}^{(n)} - \frac{i}{2} \Omega_{\mu\nu}^{(n)}$.
- **Parity-Odd Shear Metric:** $g_{xy}(W_b) = -0.252513$, $g_{xy}(-W_b) = +0.252513$, $g_{xy}(W_a) = 0$.
- **Chiral Floquet Drive:** 5-step time-periodic bond modulation stabilizes anomalous π-quasienergy HONHSE corner modes ($CD_{\pm 2} = -47.20\%$).

Slide 6: Chair44 Aperiodic Inflation & Bispectrum Chirality Audit

- **Geometric Big Bang Conjecture:** Early-universe inflation driven by discrete substitution inflation $\boldsymbol{N}(n+1) = \mathbf{M}_{\mathrm{sub}}\boldsymbol{N}(n)$ with $\lambda_{\mathrm{PF}} = \tau^3 \approx 4.236$; topological phason defect relaxation drives reheating.
- **Audit Resolution (A_{amp} vs. Bispectrum):** Single-point FFT phase $A_{\mathrm{amp}}(\boldsymbol{k})$ suffers from $\boldsymbol{r}_0$ origin-shift artifacts ($e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_0}$).
- **Translation-Invariant Proof:** The 3-point bispectrum $B(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3)$ on closed triangles $\boldsymbol{k}_1 + \boldsymbol{k}_2 + \boldsymbol{k}_3 = \mathbf{0}$ is strictly origin-independent ($\Delta B/|B| < 3 \times 10^{-16}$).

Slide 7: Empirical Multi-Messenger Cosmological Signatures

1. **CMB Birefringence:** Chern-Simons phason coupling $\theta F\tilde{F}$ yields $\beta_0 \approx 0.35^\circ$ and I_h-locked multipole spikes at $\ell = 6, 10, 12$ (absent at $\ell = 1..5$).
2. **Corner Polarization Hotspots:** Cosmological HONHSE funnels Stokes $Q/U/V$ polarization into 12 icosahedral sky vertices ($\theta_{\text{spot}} \approx 2.4^\circ$).
3. **Phasonic Magnetogenesis:** Helical Q_8 defect currents seed $B_{\text{IGM}} \sim 1.4 \times 10^{-15}\,\text{G}$ at $\lambda_B \sim 1\,\text{Mpc}$ ($f_{\text{helical}} \to 1$).
4. **FRB Golden-Ratio Multiplets:** Coherent Bragg dispersion splits Fast Radio Bursts into micro-echo multiplets with $\Delta t_{m+1}/\Delta t_m = \tau \approx 1.618$.